Social behavior through repost trees

Some people repost directly from the original post. Others pick up the message from an earlier repost. Connecting those steps shows how the message spread.

20,648	0.794	49.1%
Cascade graphs	Event classification macro F1	Repost source Hits@5

Describe the spread

How often is the original post shared directly? How many steps does the message travel?

Classify the event

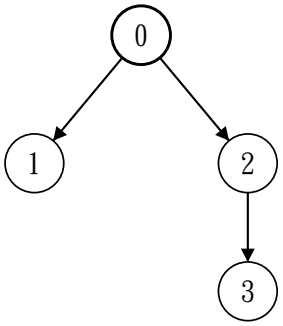
Use the complete tree to classify a cascade as normal or bursty.

Predict the repost source

Rank possible sources for a new repost using the portion of the tree already observed.

Read a repost tree

A node is a post. A directed edge points from the source post to the repost.



Post 0 is the original. Posts 1 and 2 repost it, and post 3 reposts post 2.

edge_index	The rows are [0, 0, 2] and [1, 2, 3].
x	Each post has a vector of 32 numbers.
label	The graph label is 0 for normal or 1 for burst.

Dataset splits

Split	Graphs	Normal	Burst
Training	14,453	9,633	4,820
Validation	3,097	2,064	1,033
Test	3,098	2,065	1,033

Average shape in the training set

Graph feature	Normal	Burst
Number of posts	42.44	42.70
Maximum depth	4.90	4.10
Share of posts on the widest level	39.9%	51.1%
Share of reposts directly from the original	38.2%	51.6%
Share of posts with no downstream reposts	63.0%	69.5%

The two classes are similar in size. Burst cascades have more direct reposts of the original and a greater share of posts on the widest level. Each tree has N-1 edges, so its directed density is 1/N and reflects only its size.

Classify the whole tree

Logistic regression turns a graph into a single row of numbers. Seven features describe its shape. Averaging the 32-dimensional vectors across all posts gives a second description. Combining the two produces 39 inputs.

The graph isomorphism network, or GIN, lets neighboring posts exchange information twice. It averages the resulting representations and scores the whole tree. The validation set selects the training epoch. The test set measures performance.

Test set 3,098 graphs

Method	Accuracy	Macro F1	Burst recall	ROC-AUC
Always predict normal	0.667	0.400	0.000	0.500
Post count only	0.667	0.400	0.000	0.488
Graph structure	0.731	0.664	0.428	0.766
Mean post vector	0.777	0.733	0.556	0.835
Structure and mean vector	0.822	0.794	0.674	0.886
GIN	0.796	0.760	0.612	0.850

Accuracy counts correct predictions across all samples. Macro F1 gives both classes equal weight by averaging their F1 scores. Predicting normal every time misses every burst cascade.

Where the best model made mistakes

Actual class	Predicted normal	Predicted burst
Normal	1,852	213
Burst	337	696

Combining structure and mean vectors caught 696 of 1,033 burst cascades, for a burst recall of 67.4%.

Where did a new repost come from?

The model searches within one tree. Each prediction uses only the posts and edges visible at that point.

Observe

Start with the earliest 70% of posts, rounded up, and their edges.

Rank

When post 8 arrives, posts 0 through 7 are candidates. Rank them using features such as observed out-degree, depth, and arrival order.

Reveal

Record the true source's rank. Reveal the edge, then move on to the next post.

Logistic regression trains on each true source and up to eight randomly sampled negatives. At test time, it ranks every previously observed post.

Test set 37,936 rankings

Method	Hits@1	Hits@5	MRR
Uniform random	2.8%	14.1%	0.115
Most recent first	2.6%	12.2%	0.101
Highest out-degree first	34.7%	47.0%	0.423
Logistic regression	35.2%	49.1%	0.434

Hits@5 counts how often the true source is in the top five. Mean reciprocal rank, or MRR, gives rank 1 a score of 1 and rank 2 a score of 1/2, then averages over predictions.

When the true source was a repost, logistic regression achieved 0% Hits@1 and 21.5% Hits@5 across 24,600 predictions. Every first-place hit came from an original post.

Run the experiments

Start in the repository root with Python 3.10 or later.

```
cd experiments/weibo_cascades
python -m pip install -r requirements.txt
python generate_dataset.py --out data
python run_experiments.py
```

If you already have the three .pt files, skip generation and pass --data-dir followed by the folder containing the files.

Find the outputs

File	What it contains
metrics.json	Metrics and training history
event_test_predictions.csv	Event labels and model scores
link_test_predictions.csv	True source, top choice, and rank
gin_model.pt	Model weights selected on validation

How to interpret the results

The scores describe performance in a simulation. Predicting real user interactions requires consistent user IDs across events and timestamps. Evaluating early event warnings also requires a clear cutoff so the model sees only information available before that time.